```python
 # importing required libraries.
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.preprocessing import LabelEncoder
from sklearn.model_selection import train_test_split

import warnings
warnings.filterwarnings("ignore")

# Loading the dataframe
data = pd.read_csv("Superstore.csv",encoding='latin-1')

# Displaying random records in a dataframe.
data.sample(5)
```

```
      Row ID        Order ID  Order Date  Ship Date       Ship Mode  \
9945    9946  CA-2014-111157    3/2/2014   3/6/2014  Standard Class
2367    2368  CA-2017-123659   2/10/2017  2/13/2017     First Class
766      767  CA-2014-162775   1/13/2014  1/15/2014    Second Class
4154    4155  CA-2016-139689  12/29/2016   1/3/2017  Standard Class
415      416  CA-2017-142636   11/3/2017  11/7/2017  Standard Class

     Customer ID    Customer Name    Segment        Country
City  \
9945    NH-18610    Nicole Hansen  Corporate  United States
Philadelphia
2367    MN-17935   Michael Nguyen   Consumer  United States
Clinton
766     CS-12250  Chris Selesnick  Corporate  United States  Bossier
City
4154    MP-17965    Michael Paige  Corporate  United States
Bristol
415     KC-16675  Kimberly Carter  Corporate  United States
Seattle

      ... Postal Code  Region       Product ID         Category Sub-
Category  \
9945  ...       19120    East  OFF-PA-10000327  Office Supplies
Paper
2367  ...       20735    East  OFF-PA-10002464  Office Supplies
Paper
766   ...       71111   South  OFF-BI-10004187  Office Supplies
Binders
4154  ...       37620   South  OFF-BI-10004781  Office Supplies
Binders
415   ...       98105    West  FUR-CH-10001891        Furniture
Chairs
```

```
                                          Product Name    Sales
Quantity  \
9945                                        Xerox 1971    3.424
1
2367  HP Office Recycled Paper (20Lb. and 87 Bright)   23.120
4
766                                 3-ring staple pack    5.640
3
4154                          GBC Wire Binding Strips   38.088
4
415                Global Deluxe Office Fabric Chairs  307.136
4

      Discount   Profit
9945       0.2   1.0700
2367       0.0  11.3288
766        0.0   2.7072
4154       0.7 -27.9312
415        0.2  26.8744

[5 rows x 21 columns]
```

```
# shape of the dataframe
data.shape
```

```
(9994, 21)
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 21 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   object
 3   Ship Date      9994 non-null   object
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
```

```
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
dtypes: float64(3), int64(3), object(15)
memory usage: 1.6+ MB
```

```
# Descriptive statistics for numerical coulumns.
data.describe().T
```

```
               count          mean           std       min          25%
\
Row ID        9994.0   4997.500000   2885.163629     1.000   2499.25000

Postal Code   9994.0  55190.379428  32063.693350  1040.000  23223.00000

Sales         9994.0    229.858001    623.245101     0.444     17.28000

Quantity      9994.0      3.789574      2.225110     1.000      2.00000

Discount      9994.0      0.156203      0.206452     0.000      0.00000

Profit        9994.0     28.656896    234.260108 -6599.978      1.72875


                     50%        75%        max
Row ID        4997.5000   7495.750   9994.000
Postal Code  56430.5000  90008.000  99301.000
Sales           54.4900    209.940  22638.480
Quantity         3.0000      5.000     14.000
Discount         0.2000      0.200      0.800
Profit           8.6665     29.364   8399.976
```

```
data.describe(include="all").T
```

```
                 count unique              top  freq          mean  \
Row ID          9994.0    NaN              NaN   NaN        4997.5
Order ID          9994   5009   CA-2017-100111    14           NaN
Order Date        9994   1237         9/5/2016    38           NaN
Ship Date         9994   1334       12/16/2015    35           NaN
Ship Mode         9994      4   Standard Class  5968           NaN
Customer ID       9994    793         WB-21850    37           NaN
Customer Name     9994    793    William Brown    37           NaN
Segment           9994      3         Consumer  5191           NaN
Country           9994      1    United States  9994           NaN
City              9994    531    New York City   915           NaN
State             9994     49       California  2001           NaN
Postal Code     9994.0    NaN              NaN   NaN  55190.379428
Region            9994      4             West  3203           NaN
Product ID        9994   1862  OFF-PA-10001970    19           NaN
```

```
Category         9994      3  Office Supplies  6026           NaN
Sub-Category     9994     17          Binders  1523           NaN
Product Name     9994   1850  Staple envelope    48           NaN
Sales          9994.0    NaN              NaN   NaN    229.858001
Quantity       9994.0    NaN              NaN   NaN      3.789574
Discount       9994.0    NaN              NaN   NaN      0.156203
Profit         9994.0    NaN              NaN   NaN     28.656896

                       std       min      25%      50%      75%
max
Row ID         2885.163629       1.0  2499.25   4997.5  7495.75
9994.0
Order ID               NaN       NaN      NaN      NaN      NaN
NaN
Order Date             NaN       NaN      NaN      NaN      NaN
NaN
Ship Date              NaN       NaN      NaN      NaN      NaN
NaN
Ship Mode              NaN       NaN      NaN      NaN      NaN
NaN
Customer ID            NaN       NaN      NaN      NaN      NaN
NaN
Customer Name          NaN       NaN      NaN      NaN      NaN
NaN
Segment                NaN       NaN      NaN      NaN      NaN
NaN
Country                NaN       NaN      NaN      NaN      NaN
NaN
City                   NaN       NaN      NaN      NaN      NaN
NaN
State                  NaN       NaN      NaN      NaN      NaN
NaN
Postal Code    32063.69335    1040.0  23223.0  56430.5  90008.0
99301.0
Region                 NaN       NaN      NaN      NaN      NaN
NaN
Product ID             NaN       NaN      NaN      NaN      NaN
NaN
Category               NaN       NaN      NaN      NaN      NaN
NaN
Sub-Category           NaN       NaN      NaN      NaN      NaN
NaN
Product Name           NaN       NaN      NaN      NaN      NaN
NaN
Sales           623.245101     0.444    17.28    54.49   209.94
22638.48
Quantity           2.22511       1.0      2.0      3.0      5.0
14.0
Discount          0.206452       0.0      0.0      0.2      0.2
```

```
0.8
Profit          234.260108 -6599.978  1.72875   8.6665   29.364
8399.976
```

```
data["Order Date"] = pd.to_datetime(data["Order Date"])

data["Ship Date"] = pd.to_datetime(data["Ship Date"])

data["order_month"] = pd.to_datetime(data["Order
Date"]).dt.month_name()

data["order_year"] = pd.to_datetime(data["Order Date"]).dt.year

data["Ship_month"] = pd.to_datetime(data["Ship Date"]).dt.month_name()

data["Ship_year"] = pd.to_datetime(data["Ship Date"]).dt.month_name()

data["Ship_year"] = pd.to_datetime(data["Ship Date"]).dt.year

data.sample(n=1)
```

```
      Row ID        Order ID Order Date  Ship Date       Ship Mode  \
5673    5674  CA-2017-127922 2017-10-27 2017-11-03  Standard Class

     Customer ID  Customer Name    Segment        Country    City  ...
\
5673    SH-19975  Sally Hughsby  Corporate  United States  Dallas  ...


     Sub-Category  Product Name  Sales Quantity Discount Profit
order_month  \
5673        Paper    Xerox 1972  8.448        2      0.2   2.64
October

      order_year  Ship_month  Ship_year
5673        2017    November       2017

[1 rows x 25 columns]
```

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 25 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   datetime64[ns]
 3   Ship Date      9994 non-null   datetime64[ns]
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
```

```
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
 21  order_month    9994 non-null   object
 22  order_year     9994 non-null   int32
 23  Ship_month     9994 non-null   object
 24  Ship_year      9994 non-null   int32
dtypes: datetime64[ns](2), float64(3), int32(2), int64(3), object(15)
memory usage: 1.8+ MB
```

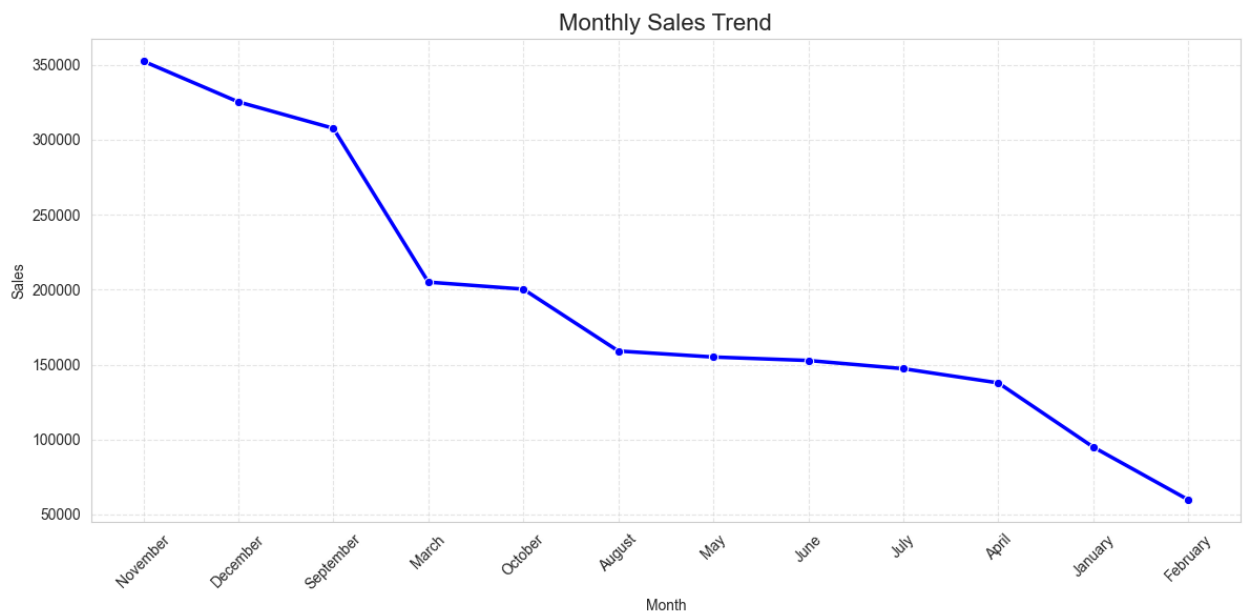
Missing Value Treatment

```
# Exploratoty Data Analysis

# Calculating Monthly Sales.
monthly_sales = data.groupby("order_month")
["Sales"].sum().reset_index().sort_values(by='Sales',ascending=False)

monthly_sales
```

```
   order_month        Sales
9     November  352461.0710
2     December  325293.5035
11   September  307649.9457
7        March  205005.4888
10     October  200322.9847
1       August  159044.0630
8          May  155028.8117
6         June  152718.6793
5         July  147238.0970
0        April  137762.1286
4      January   94924.8356
3     February   59751.2514
```

```
plt.figure(figsize=(12,6))
sns.set_style("whitegrid")
```

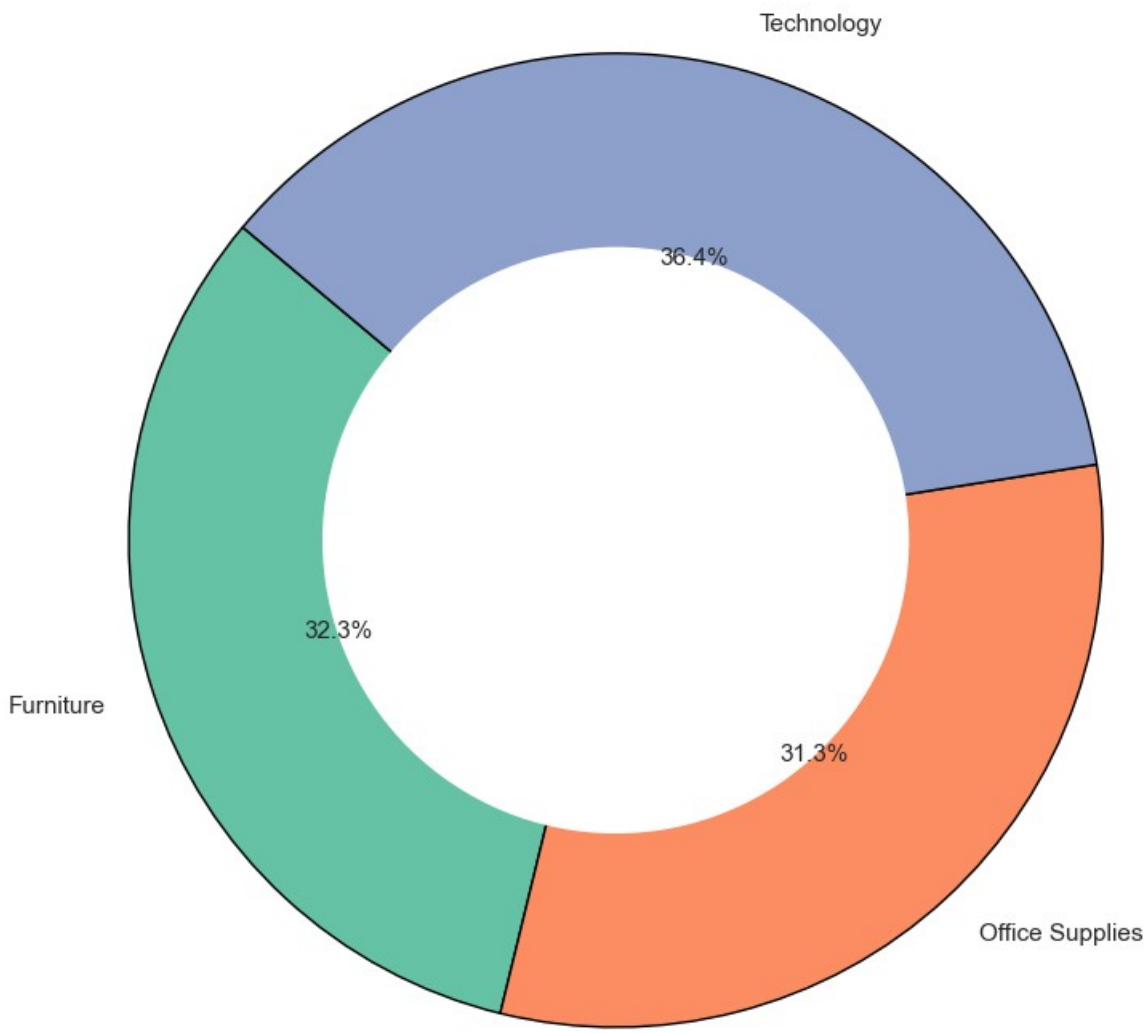
```
sns.lineplot(
    monthly_sales,
    x="order_month",
    y="Sales",
    marker="o",
    linewidth=2.5,
    color="blue"
)

plt.title("Monthly Sales Trend", fontsize=16)
plt.xlabel("Month")
plt.ylabel("Sales")

plt.xticks(rotation=45)

plt.grid(True, linestyle="--", alpha=0.5)

plt.tight_layout()
plt.show()
```




```
# Sales on the basis of Category
category_sales = data.groupby('Category')["Sales"].sum()

category_sales
```

```
Category
Furniture           741999.7953
Office Supplies     719047.0320
Technology          836154.0330
Name: Sales, dtype: float64
```

```
plt.figure(figsize=(8,8))
sns.set_style("white")

colors = sns.color_palette("Set2")

plt.pie(
    category_sales,
    labels=category_sales.index,
    autopct="%1.1f%%",
    colors=colors,
    startangle=140,
    wedgeprops={'edgecolor': 'black', 'linewidth': 1},
    textprops={'fontsize': 11}
)

plt.title("Sales by Category", fontsize=0.0, fontweight='bold')

# donut style (modern look)
centre_circle = plt.Circle((0,0), 0.60, fc='white')
fig = plt.gcf()
fig.gca().add_artist(centre_circle)

plt.tight_layout()
plt.show()
```

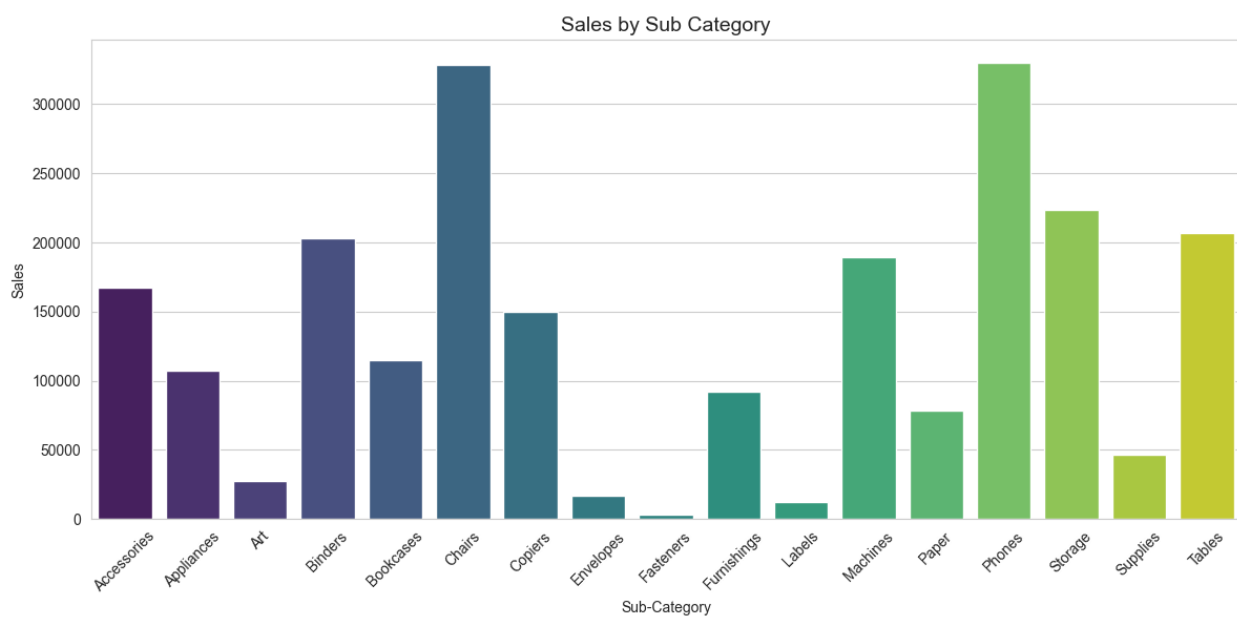



```python
# sales per Sub-Category
Sub_category_sales = data.groupby('Sub-Category')
["Sales"].sum().reset_index()

Sub_category_sales
```

```
   Sub-Category        Sales
0   Accessories  167380.3180
1    Appliances  107532.1610
2           Art   27118.7920
3       Binders  203412.7330
4     Bookcases  114879.9963
```

```
5         Chairs  328449.1030
6        Copiers  149528.0300
7      Envelopes   16476.4020
8      Fasteners    3024.2800
9    Furnishings   91705.1640
10        Labels   12486.3120
11      Machines  189238.6310
12         Paper   78479.2060
13        Phones  330007.0540
14       Storage  223843.6080
15      Supplies   46673.5380
16        Tables  206965.5320
```

```
plt.figure(figsize=(12,6))
sns.set_style("whitegrid")

sns.barplot(
    Sub_category_sales,
    x='Sub-Category',
    y='Sales',
    palette='viridis'
)

plt.title("Sales by Sub Category", fontsize=14)
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```

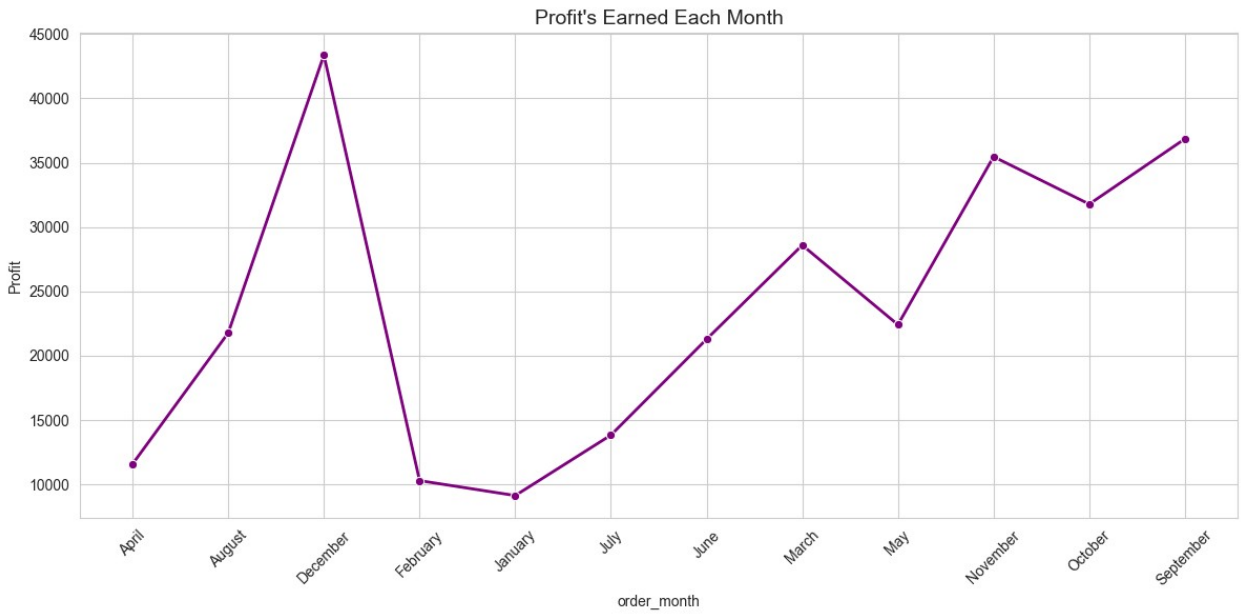


Profit Analysis

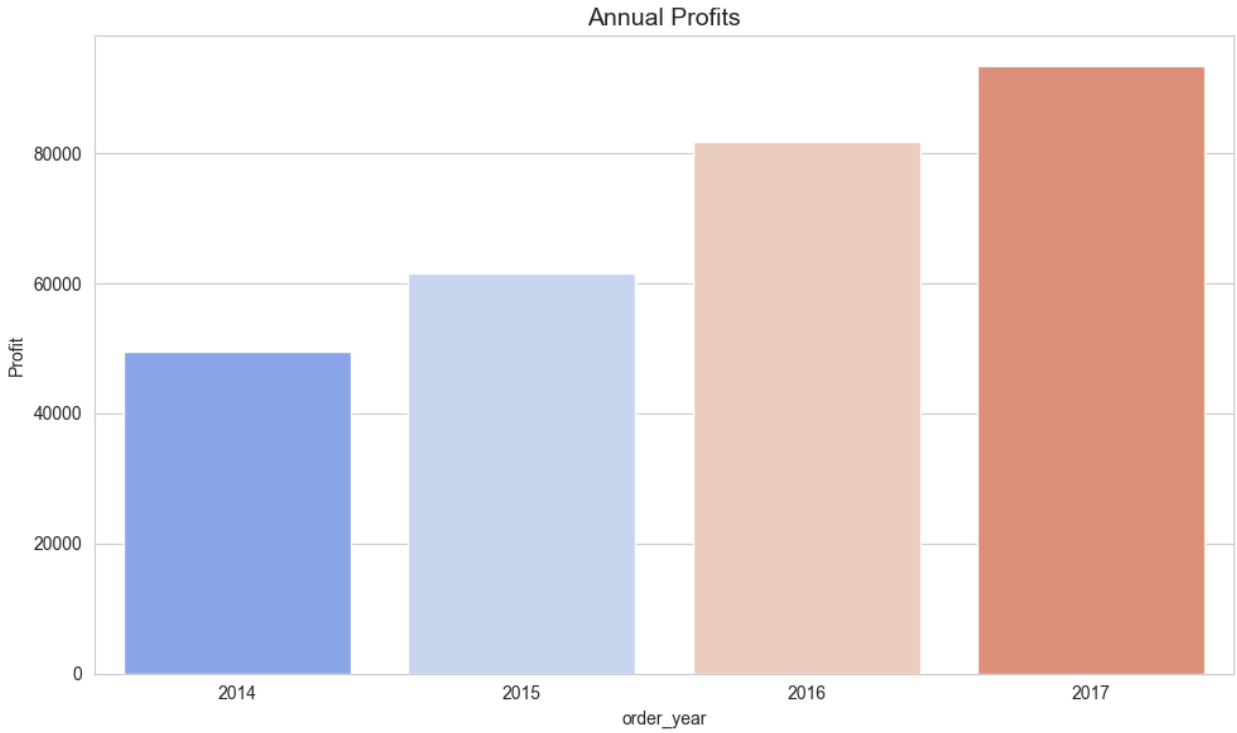
```
# Profit's earned each month.
monthly_profit = data.groupby("order_month")
["Profit"].sum().reset_index()

monthly_profit
```

```
   order_month      Profit
0        April  11587.4363
1       August  21776.9384
2     December  43369.1919
3     February  10294.6107
4      January   9134.4461
5         July  13832.6648
6         June  21285.7954
7        March  28594.6872
8          May  22411.3078
9     November  35468.4265
10     October  31784.0413
11   September  36857.4753
```

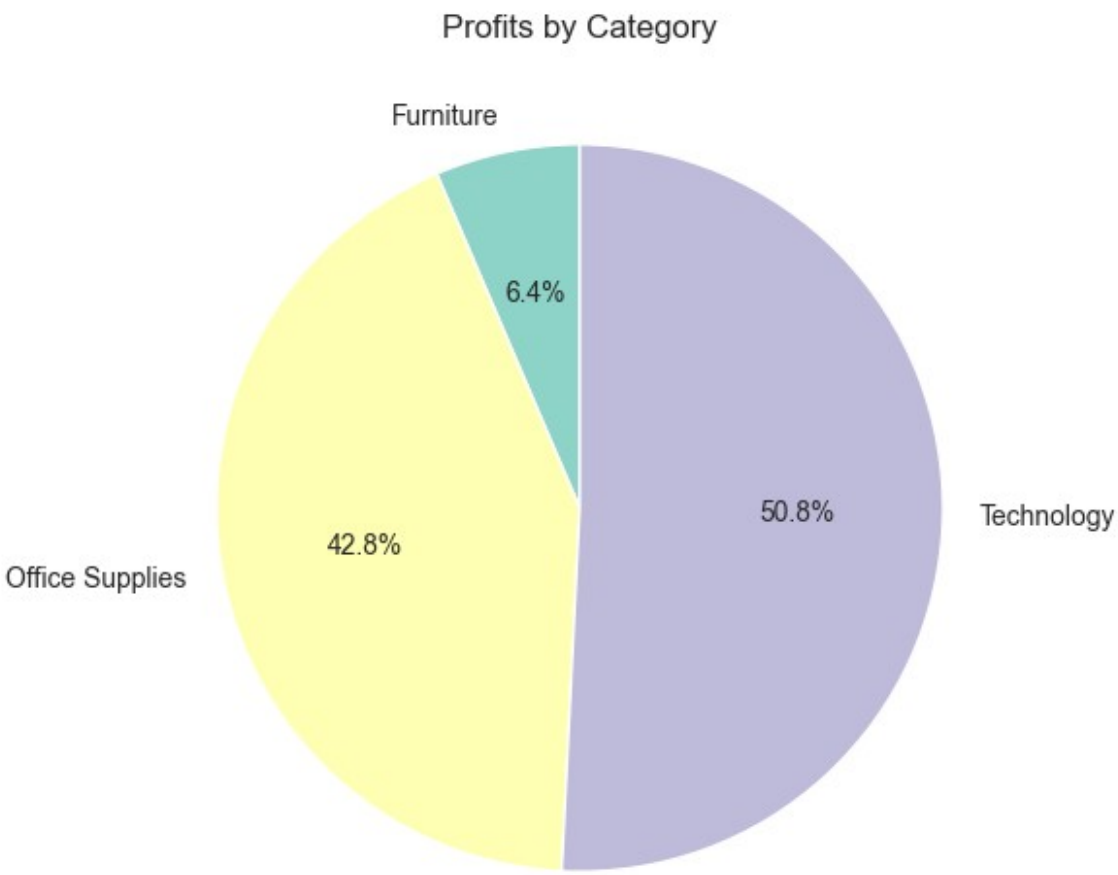
```
plt.figure(figsize=(12,6))
sns.set_style("whitegrid")

sns.lineplot(
    monthly_profit,
    x="order_month",
    y="Profit",
    marker="o",
    linewidth=2,
    color="purple"
)

plt.title("Profit's Earned Each Month", fontsize=14)
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```




```
# profit's earned per annum.
annual_profit = data.groupby("order_year")
["Profit"].sum().reset_index()

plt.figure(figsize=(10,6))
sns.set_style("whitegrid")

sns.barplot(
    annual_profit,
    x="order_year",
    y="Profit",
    palette="coolwarm"
)

plt.title("Annual Profits", fontsize=14)

plt.tight_layout()
plt.show()
```




```
#Profit's earned by each category.
category_profits = data.groupby("Category")["Profit"].sum()

plt.figure(figsize=(6,6))
sns.set_style("white")

plt.title("Profits by Category")

colors = sns.color_palette("Set3")

plt.pie(
    category_profits,
    labels=category_profits.index,
    autopct='%1.1f%%',
    colors=colors,
    startangle=90
)

plt.tight_layout()
plt.show()
```

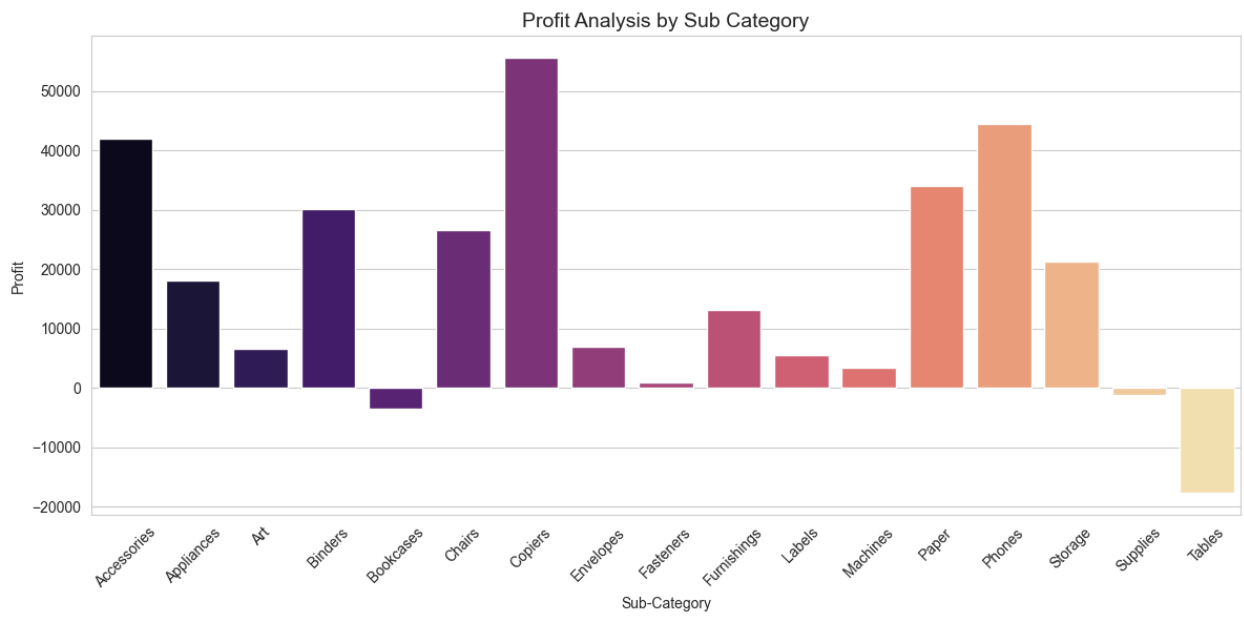



```
# Profit's earned by each sub category
sub_profit = data.groupby("Sub-Category")
["Profit"].sum().reset_index()

sub_profit
```

	Sub-Category	Profit
0	Accessories	41936.6357
1	Appliances	18138.0054
2	Art	6527.7870
3	Binders	30221.7633
4	Bookcases	-3472.5560
5	Chairs	26590.1663
6	Copiers	55617.8249
7	Envelopes	6964.1767
8	Fasteners	949.5182
9	Furnishings	13059.1436
10	Labels	5546.2540
11	Machines	3384.7569
12	Paper	34053.5693

```
13       Phones  44515.7306
14      Storage  21278.8264
15     Supplies  -1189.0995
16       Tables -17725.4811
```

```
plt.figure(figsize=(12,6))
sns.set_style("whitegrid")

sns.barplot(
    sub_profit,
    x="Sub-Category",
    y="Profit",
    palette="magma"
)

plt.title("Profit Analysis by Sub Category", fontsize=14)
plt.xticks(rotation=45)

plt.tight_layout()
plt.show()
```




```
segment = data.groupby("Segment")[["Sales","Profit"]].sum()

segment
```

```
                     Sales       Profit
Segment
Consumer     1.161401e+06  134119.2092
Corporate    7.061464e+05   91979.1340
Home Office  4.296531e+05   60298.6785
```

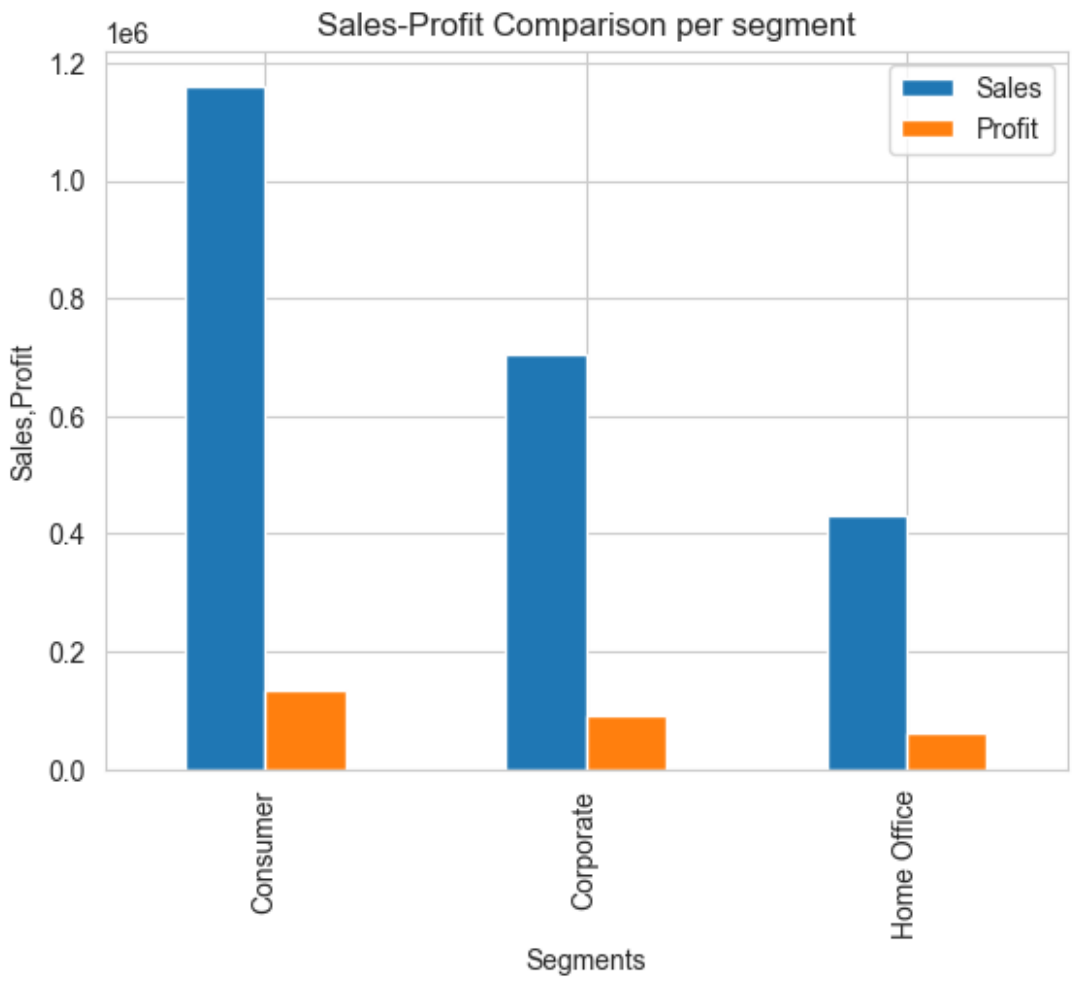
```
# Shape of the DataFrame
data.shape
```

```
(9994, 21)
```

```
plt.figure(figsize=(12, 6))
segment.plot(kind='bar')
plt.title("Sales-Profit Comparison per segment")
plt.xlabel("Segments")
plt.ylabel("Sales,Profit")
```

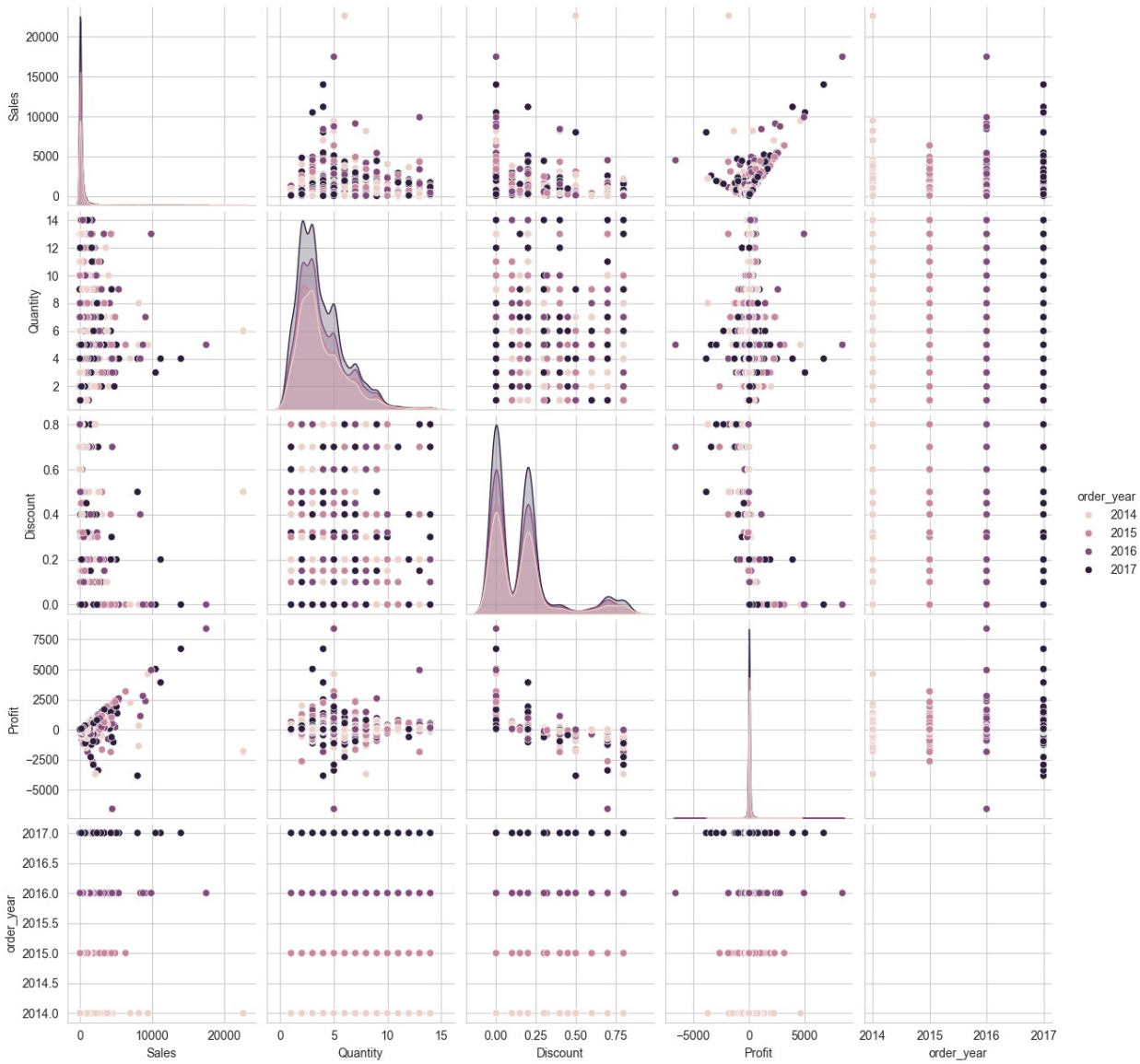
```
Text(0, 0.5, 'Sales,Profit')
```

```
<Figure size 1200x600 with 0 Axes>
```




```
data['order_year'] = pd.to_datetime(data['Order Date']).dt.year

sns.pairplot(
    data,
```

```
    vars=['Sales', 'Quantity', 'Discount', 'Profit', 'order_year'],
    hue='order_year'
)

plt.show()
```




```
# Shape of the DataFrame
data.shape
```

```
(9994, 22)
```

```
# create Ship_year column first
data['Ship_year'] = pd.to_datetime(data['Ship Date']).dt.year

# now your same code
```

```
Num1 = data[['Row ID', 'Postal Code','Sales', 'Quantity', 'Discount',
'Profit', 'Ship_year']]

# create missing columns
data['order_month'] = pd.to_datetime(data['Order
Date']).dt.to_period('M').astype(str)
data['Ship_month'] = pd.to_datetime(data['Ship
Date']).dt.to_period('M').astype(str)

# now your same code
Cat1 = data[['Order ID', 'Order Date', 'Ship Date', 'Ship Mode',
       'Customer ID', 'Customer Name', 'Segment', 'Country', 'City',
'State',
       'Region', 'Product ID', 'Category', 'Sub-Category',
'order_month', 'Ship_month',
       'Product Name']]

columns = data[['Row ID', 'Sales', 'Quantity', 'Discount', 'Profit']]

columns.info()
```

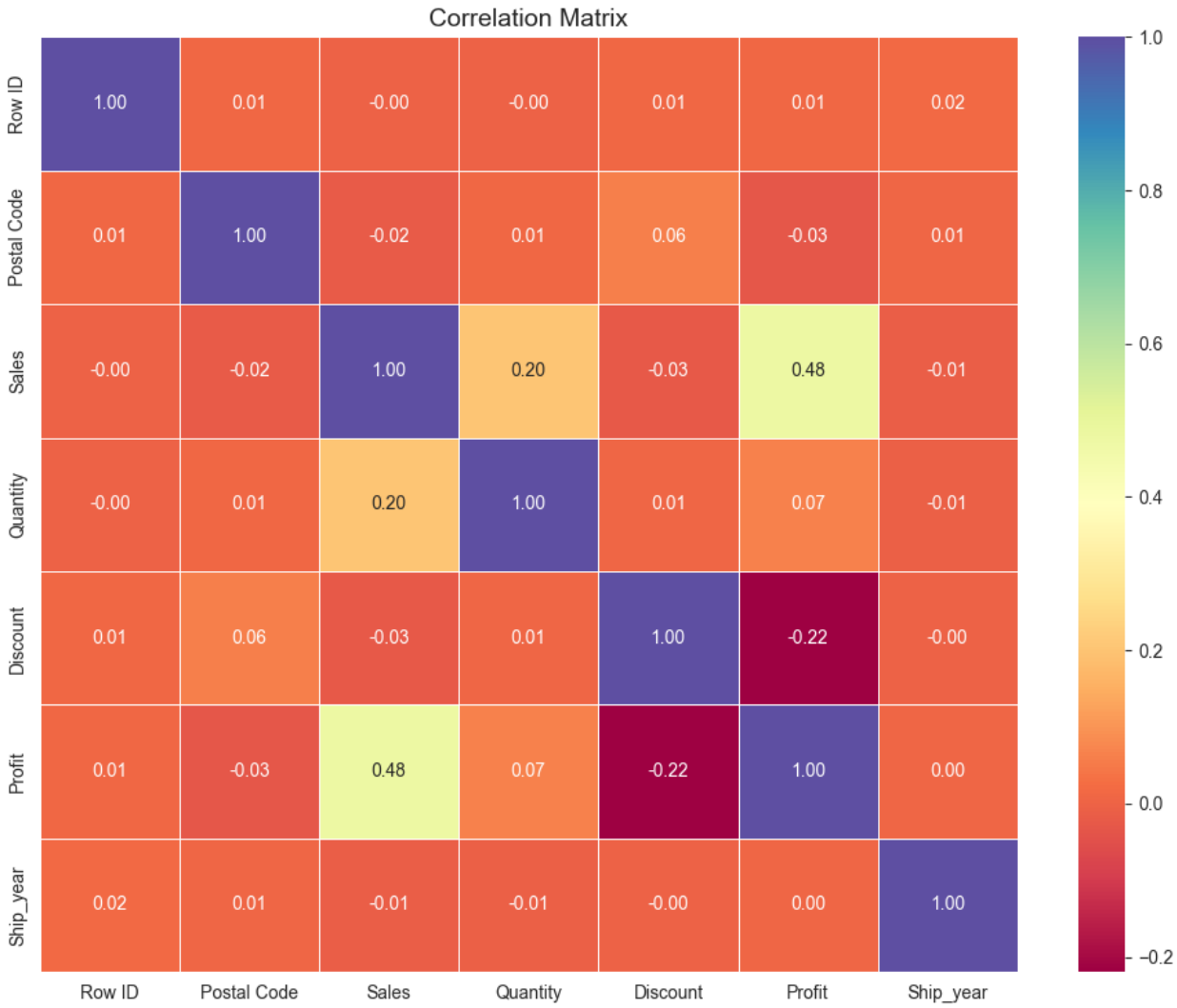
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 5 columns):
 #   Column    Non-Null Count  Dtype
---  ------    --------------  -----
 0   Row ID    9994 non-null   int64
 1   Sales     9994 non-null   float64
 2   Quantity  9994 non-null   int64
 3   Discount  9994 non-null   float64
 4   Profit    9994 non-null   float64
dtypes: float64(3), int64(2)
memory usage: 390.5 KB
```

```
data.info()
```

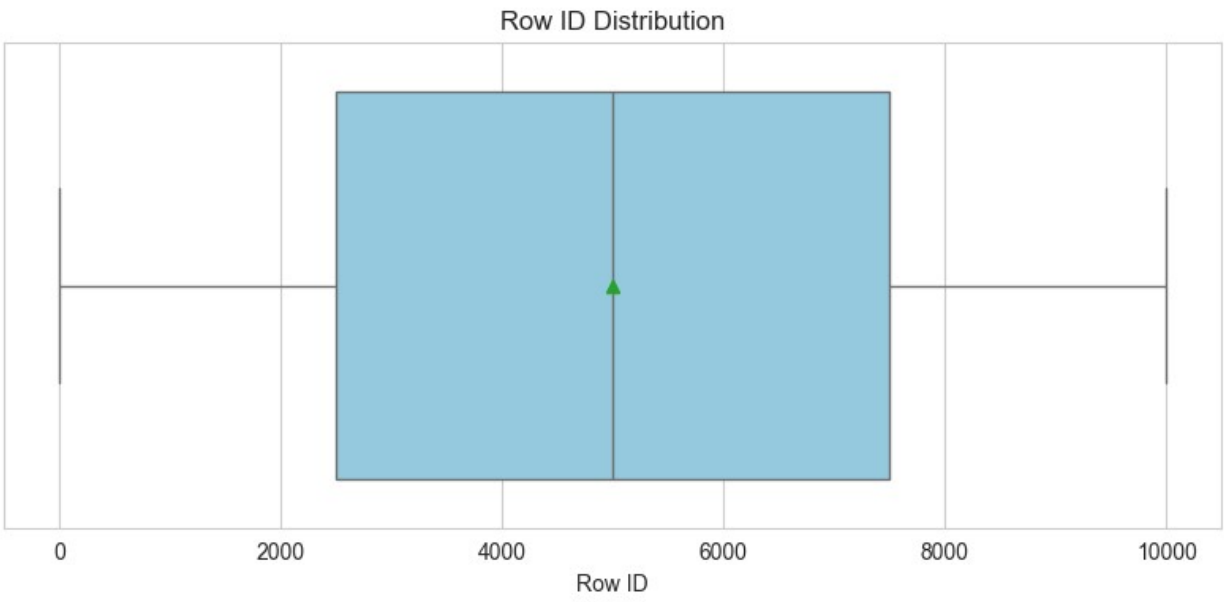
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9994 entries, 0 to 9993
Data columns (total 25 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   Row ID         9994 non-null   int64
 1   Order ID       9994 non-null   object
 2   Order Date     9994 non-null   object
 3   Ship Date      9994 non-null   object
 4   Ship Mode      9994 non-null   object
 5   Customer ID    9994 non-null   object
 6   Customer Name  9994 non-null   object
 7   Segment        9994 non-null   object
 8   Country        9994 non-null   object
 9   City           9994 non-null   object
```

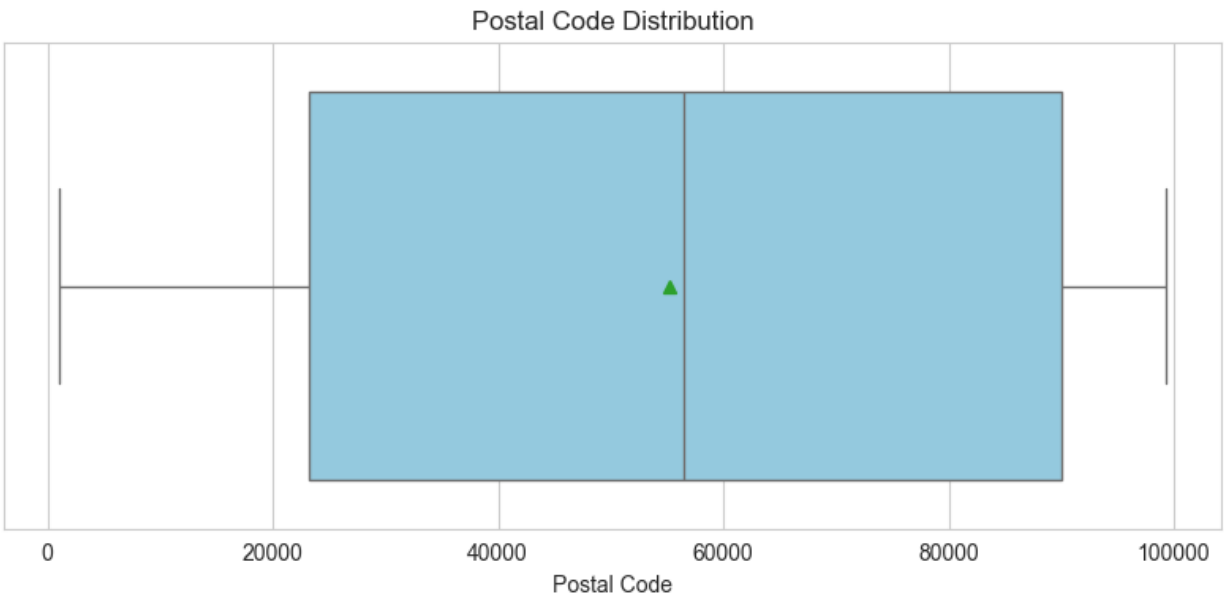
```
 10  State          9994 non-null   object
 11  Postal Code    9994 non-null   int64
 12  Region         9994 non-null   object
 13  Product ID     9994 non-null   object
 14  Category       9994 non-null   object
 15  Sub-Category   9994 non-null   object
 16  Product Name   9994 non-null   object
 17  Sales          9994 non-null   float64
 18  Quantity       9994 non-null   int64
 19  Discount       9994 non-null   float64
 20  Profit         9994 non-null   float64
 21  order_year     9994 non-null   int32
 22  Ship_year      9994 non-null   int32
 23  order_month    9994 non-null   object
 24  Ship_month     9994 non-null   object
dtypes: float64(3), int32(2), int64(3), object(17)
memory usage: 1.8+ MB
```

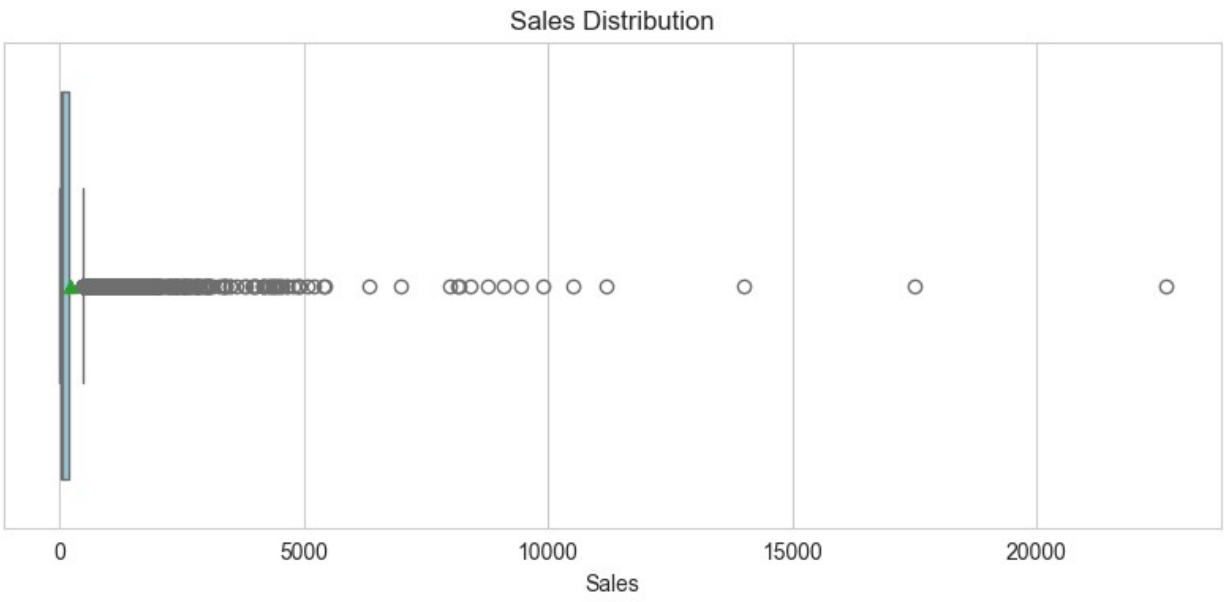
```python
# Create a correlation matrix
corr_matrix = Num1.corr()

# # Plot the correlation matrix using a heatmap
# plt.figure(figsize=(8, 8))
# sns.heatmap(corr_matrix, annot=True, cmap='coolwarm', fmt='.2f')

plt.figure(figsize=(10,8))
sns.set_style("white")

sns.heatmap(
    corr_matrix,
    annot=True,
    cmap="Spectral",
    fmt=".2f",
    linewidths=0.5,
    linecolor='white',
    cbar=True
)

plt.title("Correlation Matrix", fontsize=14)

plt.tight_layout()
plt.show()
```

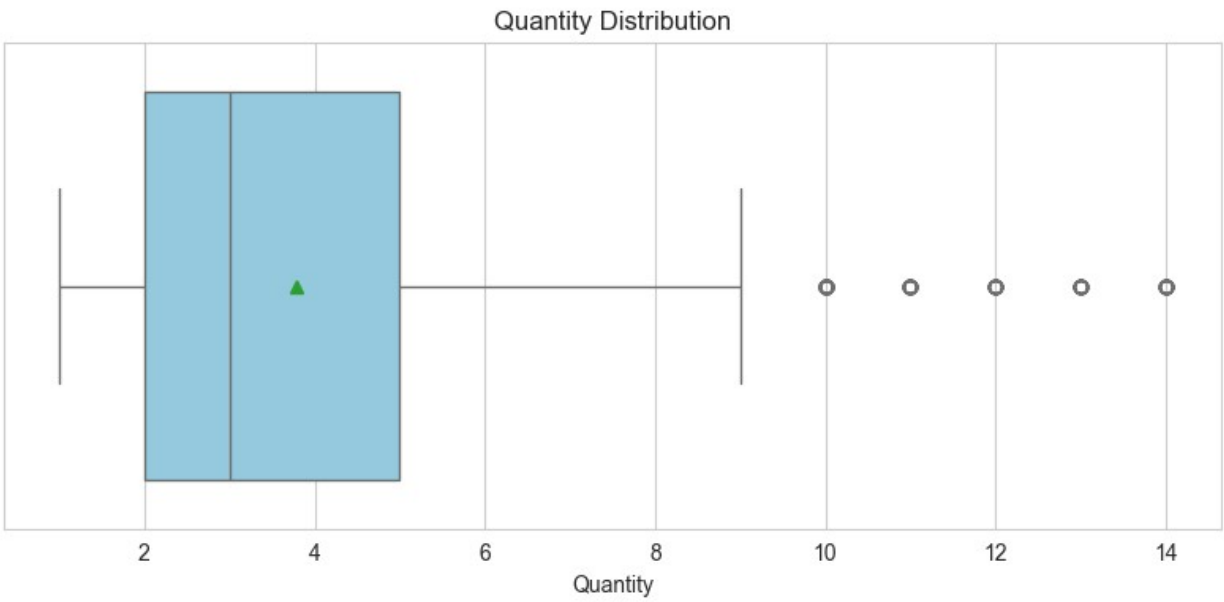


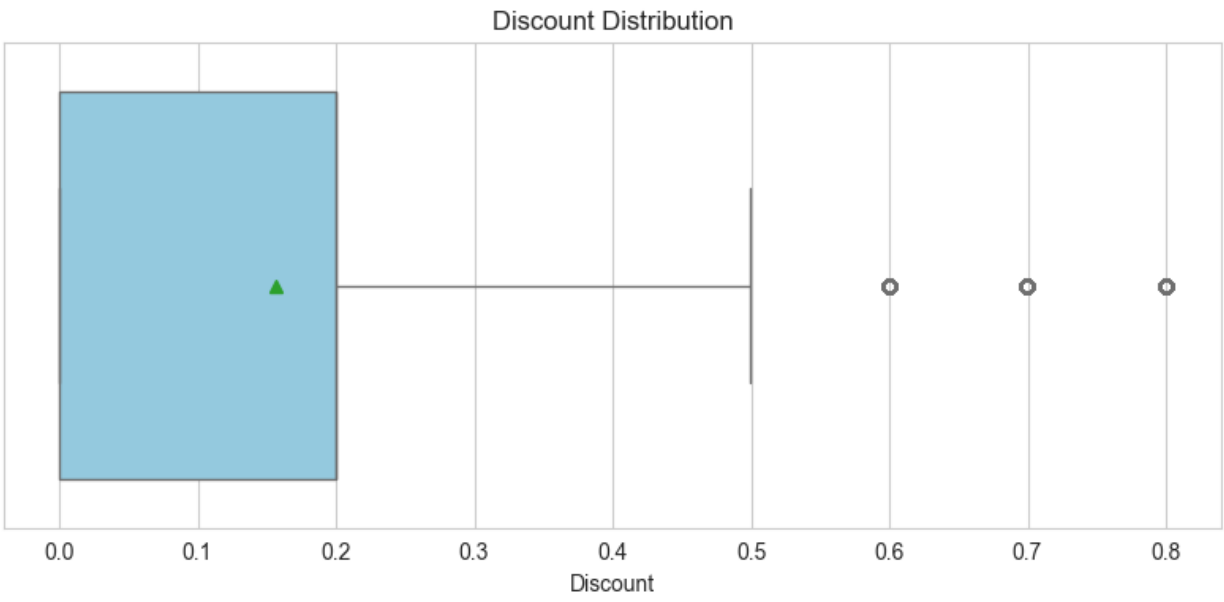

```python
sns.set_style("whitegrid")

for col in Num1.columns:
    plt.figure(figsize=(8,4))

    sns.boxplot(
        x=Num1[col],
        showmeans=True,
        color="skyblue"
    )

    plt.title(f"{col} Distribution", fontsize=12)

    plt.tight_layout()
    plt.show()
```

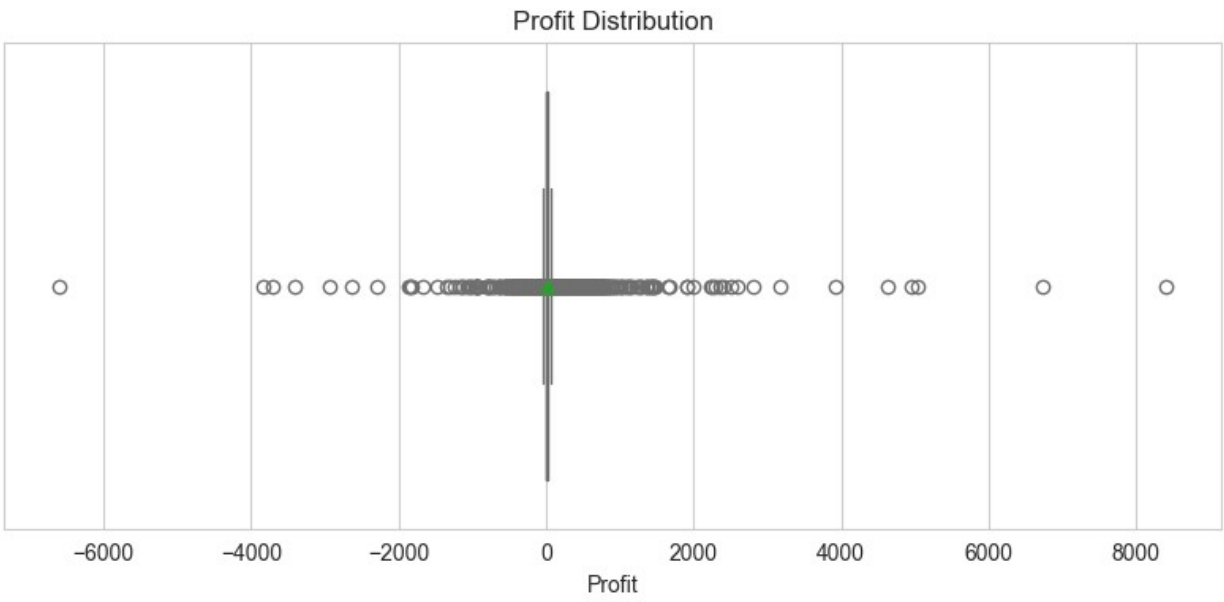

Row ID Distribution
0
2000
4000
6000
8000
10000
Row ID

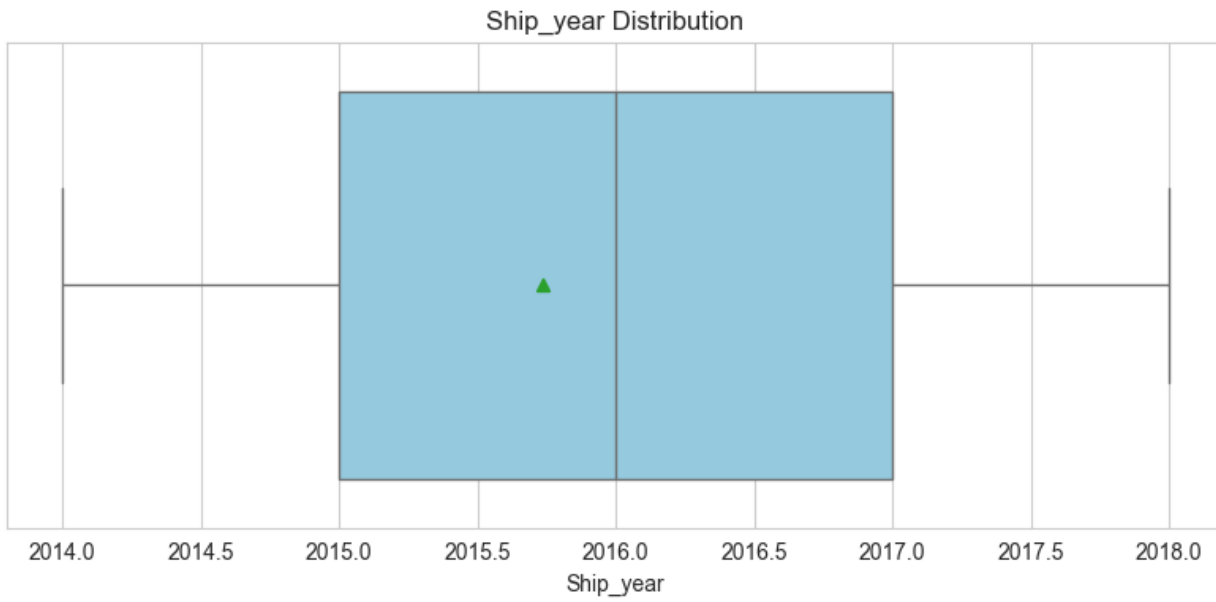

Postal Code Distribution
0
20000
40000
60000
80000
100000
Postal Code


Sales Distribution
0
5000
10000
15000
20000
Sales


Quantity Distribution
2
4
6
8
10
12
14
Quantity


Discount Distribution
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
Discount


Profit Distribution
−6000
−4000
−2000
0
2000
4000
6000
8000
Profit




```
# Using exponential method to reduce outliers
exp = data[[ 'Sales', 'Quantity', 'Discount', 'Profit']]

np.exp(exp["Sales"])
```

```
0       5.858447e+113
1                 inf
2        2.235555e+06
3                 inf
4        5.179633e+09
            ...
9989     9.227139e+10
9990     8.664042e+39
9991    1.986689e+112
9992     7.163358e+12
9993    4.009094e+105
Name: Sales, Length: 9994, dtype: float64
```

```
# Using for-loop for reducing treatment
for i in exp.columns:
    exp[i] = np.exp(exp[i])

exp
```

```
              Sales    Quantity  Discount         Profit
0     5.858447e+113    7.389056  1.000000   1.595310e+18
1               inf   20.085537  1.000000   2.308080e+95
2      2.235555e+06    7.389056  1.000000   9.642976e+02
3               inf  148.413159  1.568312  4.484875e-167
4      5.179633e+09    7.389056  1.221403   1.238393e+01
...             ...         ...       ...            ...
9989   9.227139e+10   20.085537  1.221403   6.050948e+01
```

```
9990   8.664042e+39    7.389056  1.000000   6.157614e+06
9991  1.986689e+112    7.389056  1.221403   2.644598e+08
9992   7.163358e+12   54.598150  1.000000   6.092598e+05
9993  4.009094e+105    7.389056  1.000000   4.796383e+31

[9994 rows x 4 columns]
```

```
# Replacing the values of Num1 dataframe with exponential values.
Num1[[ 'Sales', 'Quantity', 'Discount', 'Profit']] = exp[[ 'Sales',
'Quantity', 'Discount', 'Profit']]

Num1
```

```
      Row ID  Postal Code          Sales    Quantity  Discount
Profit  \
0          1        42420  5.858447e+113    7.389056  1.000000
1.595310e+18
1          2        42420            inf   20.085537  1.000000
2.308080e+95
2          3        90036   2.235555e+06    7.389056  1.000000
9.642976e+02
3          4        33311            inf  148.413159  1.568312
4.484875e-167
4          5        33311   5.179633e+09    7.389056  1.221403
1.238393e+01
...      ...          ...            ...         ...       ...
...
9989    9990        33180   9.227139e+10   20.085537  1.221403
6.050948e+01
9990    9991        92627   8.664042e+39    7.389056  1.000000
6.157614e+06
9991    9992        92627  1.986689e+112    7.389056  1.221403
2.644598e+08
9992    9993        92627   7.163358e+12   54.598150  1.000000
6.092598e+05
9993    9994        92683  4.009094e+105    7.389056  1.000000
4.796383e+31

      Ship_year
0          2016
1          2016
2          2016
3          2015
4          2015
...         ...
9989       2014
9990       2017
9991       2017
9992       2017
9993       2017
```

```
[9994 rows x 7 columns]
```